

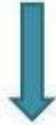
Module 4: Particle dynamics I

Newton's laws of motion

Newton's First Law

Forces are balanced

Resultant force (ΣF) = 0 N



acceleration (a) = 0 m s⁻²

} What happens to acceleration when forces are balanced?



Objects at rest

($V = 0$ m/s)



Stay at rest

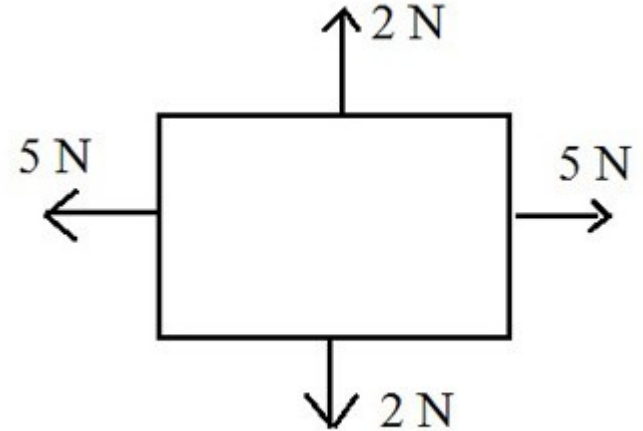
What happens to the motion of an object when acceleration is zero?

Objects in uniform motion

($V = \text{constant} \neq 0$ m/s)



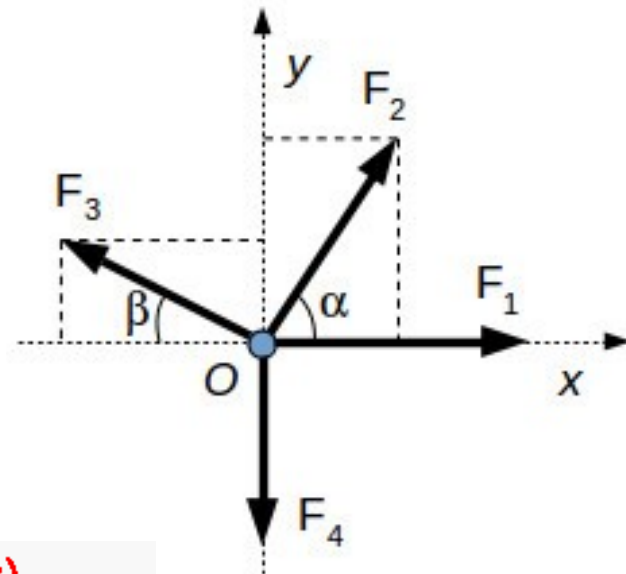
Stay in uniform motion
Same speed & direction



Newton's laws of motion

Newton's second law of motion

$$\vec{F}_{\text{Net}} = m\vec{a} = \begin{cases} \vec{F}_{\text{Net}_x} = m\vec{a}_x \\ \vec{F}_{\text{Net}_y} = m\vec{a}_y \\ \vec{F}_{\text{Net}_z} = m\vec{a}_z \end{cases} \text{ where } \begin{cases} \vec{F}_{\text{Net}_x} = \sum_i \vec{F}_{i_x} \\ \vec{F}_{\text{Net}_y} = \sum_i \vec{F}_{i_y} \\ \vec{F}_{\text{Net}_z} = \sum_i \vec{F}_{i_z} \end{cases}$$



Differential Form: Force = change of momentum
with change of time

$$F = \frac{d(mv)}{dt}$$

or:

Force = change in mass X velocity with time

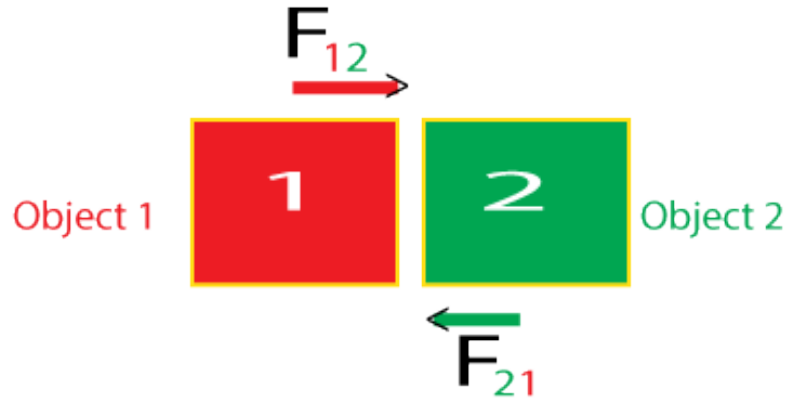
$$F = \frac{(m_1 V_1 - m_0 V_0)}{(t_1 - t_0)}$$

With mass constant: Force = mass X acceleration

$$F = m a$$

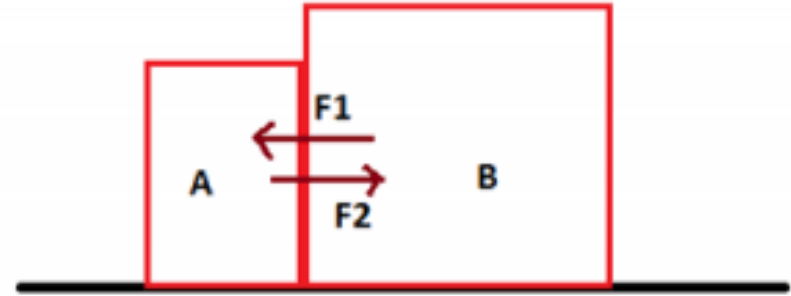
Newton's laws of motion

Newton's third law of motion



Therefore,

$$F_{12} = F_{21}$$



Work, power and impulse

$$W = \int \mathbf{F} \cdot d\mathbf{r}$$

If $\mathbf{F}$ is force and $d\mathbf{r}$ is the displacement, then W is the work done

$$P = \frac{dW}{dt}$$

P is the power

Impulse, I , is defined as

$$I = \int_{t_1}^{t_2} \mathbf{F} \cdot dt = m\mathbf{v}_2 - m\mathbf{v}_1$$

Conservative force

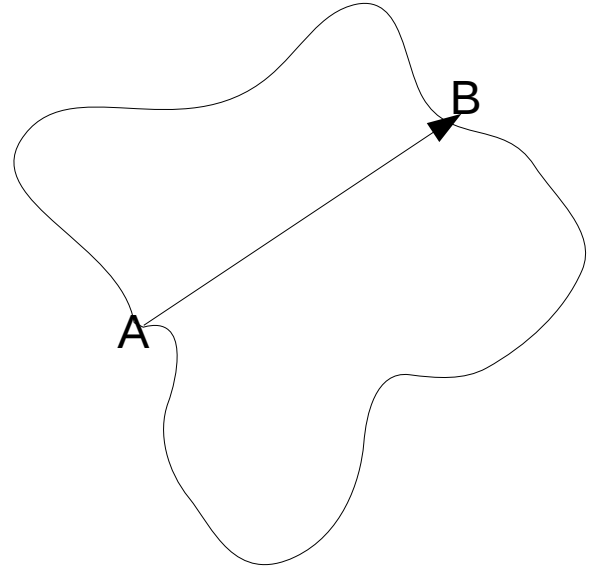
A force is said to be conservative if, work done by the force is independent of the path taken

$$W = \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r} = V(P_2) - V(P_1)$$

$$W = \oint \mathbf{F} \cdot d\mathbf{r} = 0$$

$$\mathbf{F} = -\nabla V$$

$$\nabla \times \mathbf{F} = 0$$

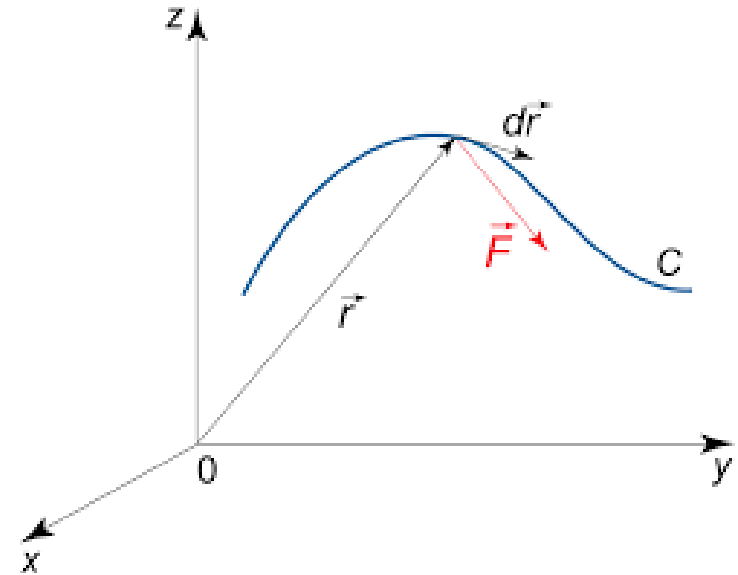


Conservation of energy

Work $W = \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r}$

Kinetic energy $W = \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r} = \frac{1}{2}m(v_2^2 - v_1^2) = T_2 - T_1$

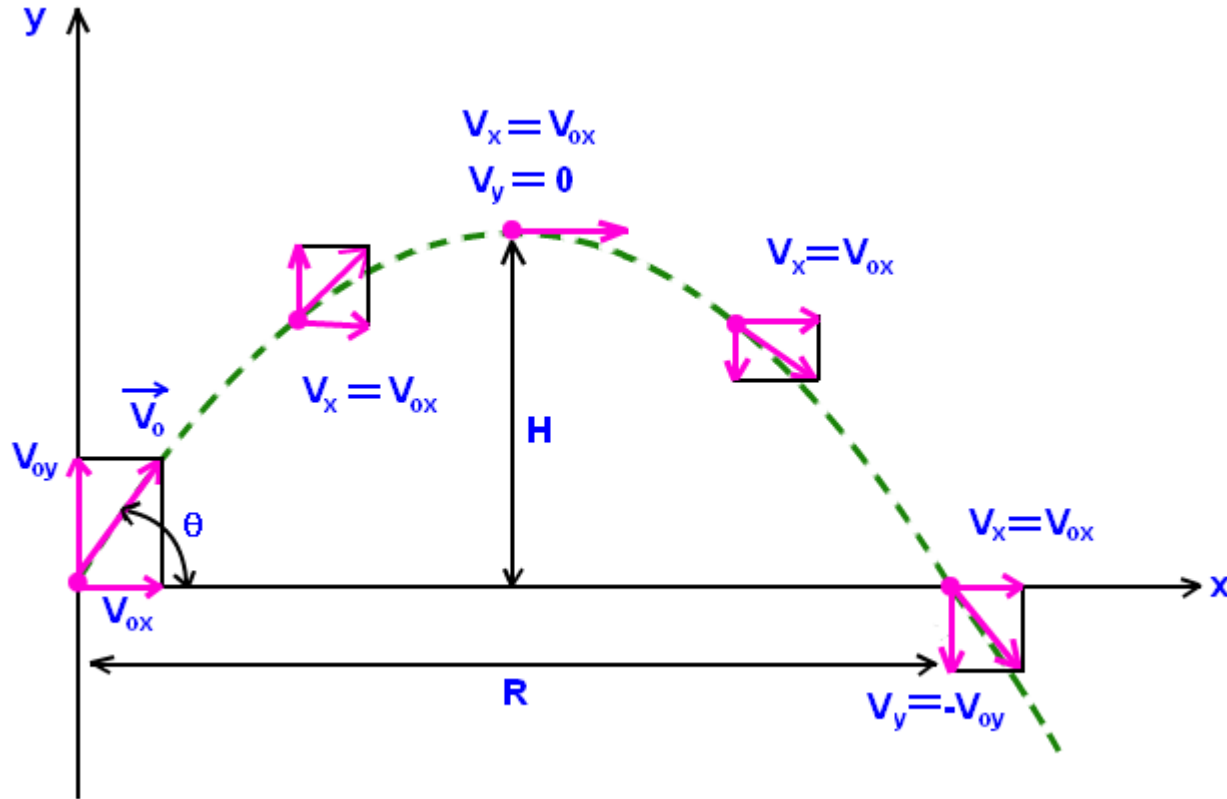
Potential energy $W = - \int_{P_1}^{P_2} \mathbf{F} \cdot d\mathbf{r} = U_1 - U_2$



For a conservative force $W = U_1 - U_2 = T_2 - T_1$

$$U_1 + T_1 = U_2 + T_2 = \text{constant}$$

Projectile motion



$$v_{ox} = v_o \cos(\theta)$$

$$v_x = v_o \cos(\theta)$$

$$v_{oy} = v_o \sin(\theta)$$

$$v_y = v_o \sin(\theta) - g t$$

Calculate the time of flight, maximum height H , range R , and relation between H and R .

Circular motion

$$v = \frac{dS}{dt} = R \frac{d\theta}{dt} = r \omega$$

$$\frac{dv}{dt} = \frac{d^2 S}{dt^2} = r \frac{d^2 \theta}{dt^2} = r \alpha$$

Angular speed

$$\omega = \frac{d\theta}{dt}$$

Angular acceleration

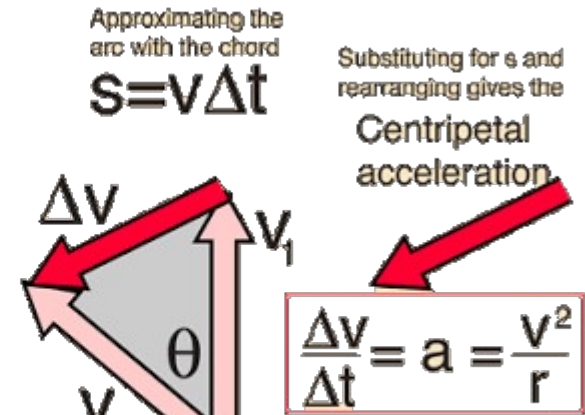
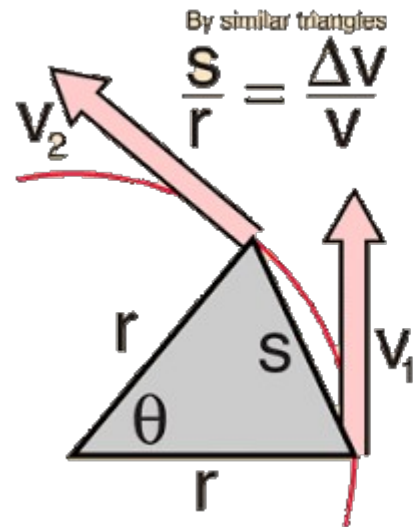
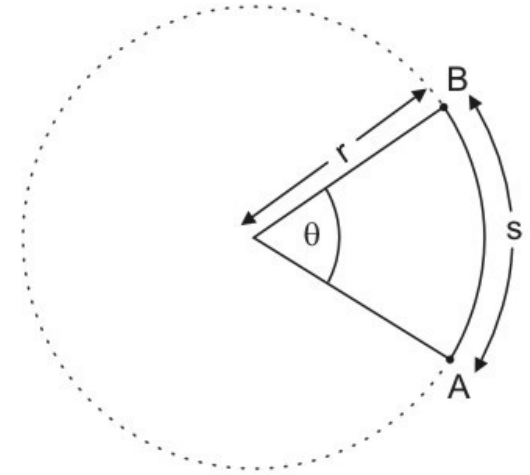
$$\frac{d^2 \theta}{dt^2} = \alpha$$

Centripetal acceleration

$$a_c = r \omega^2$$

Centripetal Force

$$F_c = ma = mr \omega^2$$



Torque

$$\mathbf{L} = \mathbf{r} \times \mathbf{F}$$

$$\mathbf{L} = m(\mathbf{r} \times \mathbf{v}) = \mathbf{r} \times \mathbf{p}$$

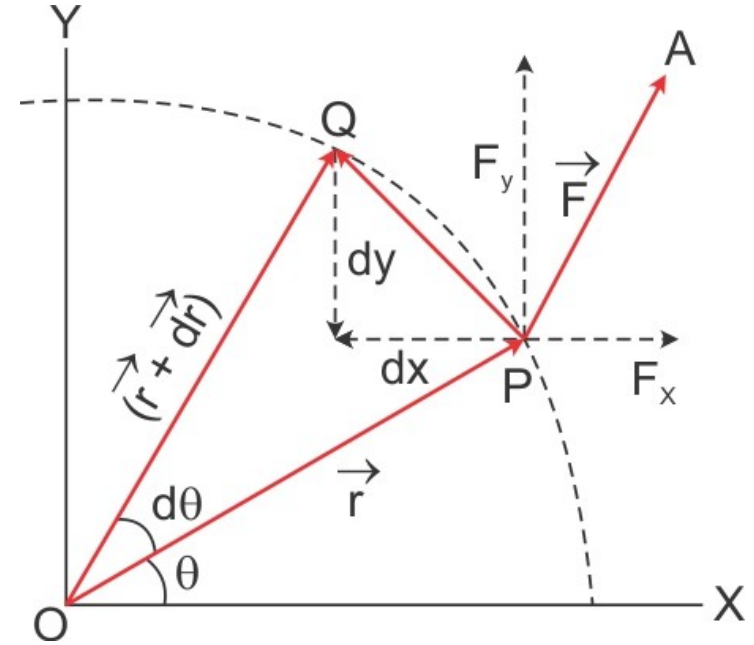
$$L = \frac{dL}{dt}$$

For $L = 0$

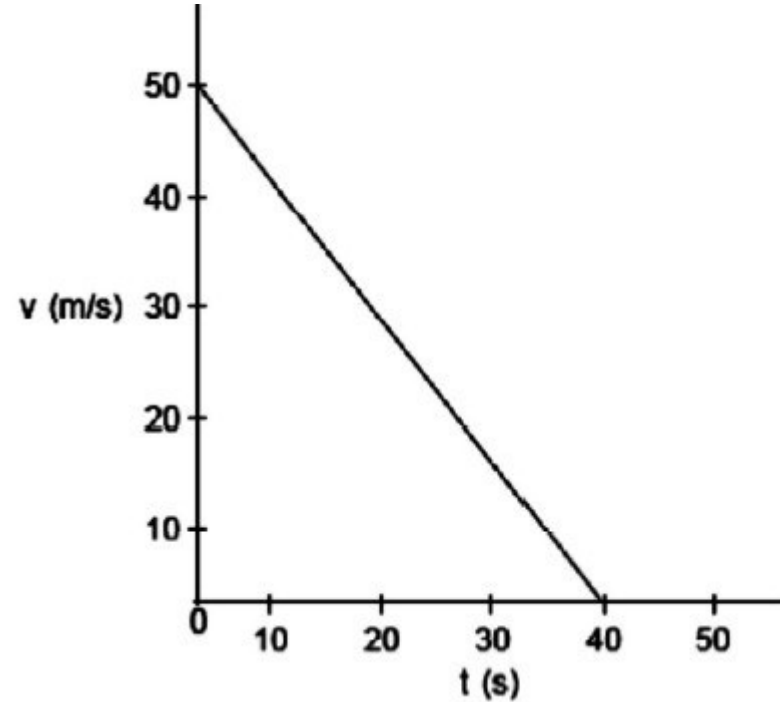
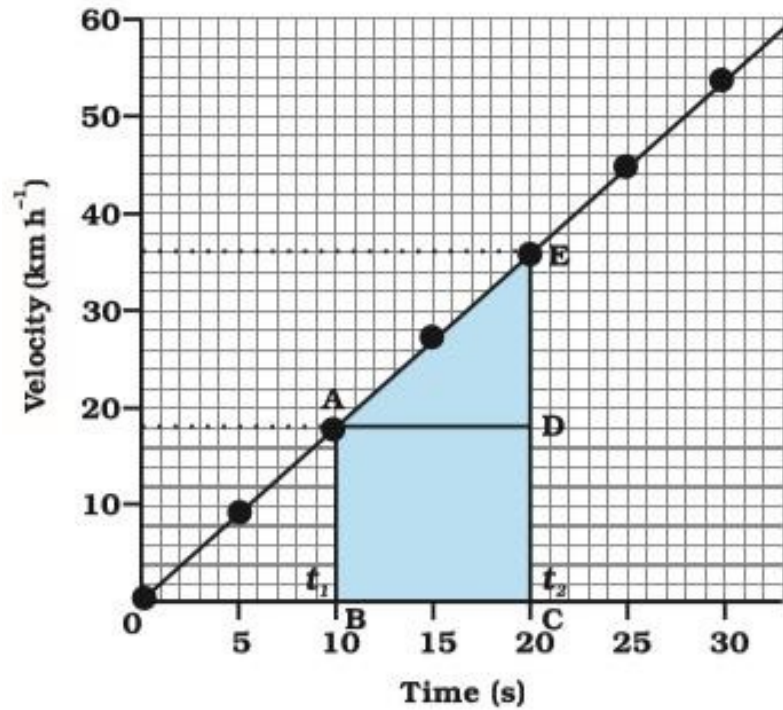
$$\frac{dL}{dt} = 0$$

$$m(\mathbf{r} \times \mathbf{v}) = \text{constant}$$

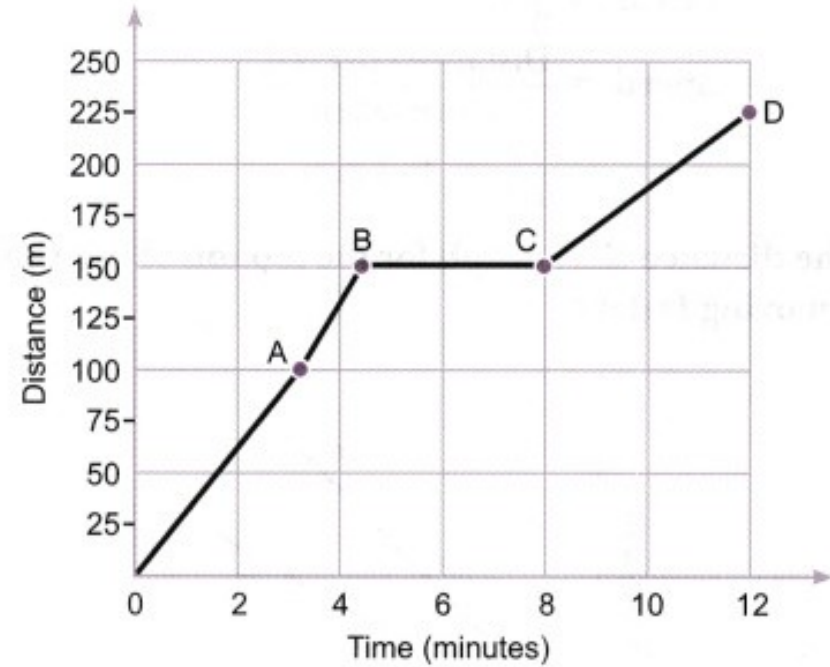
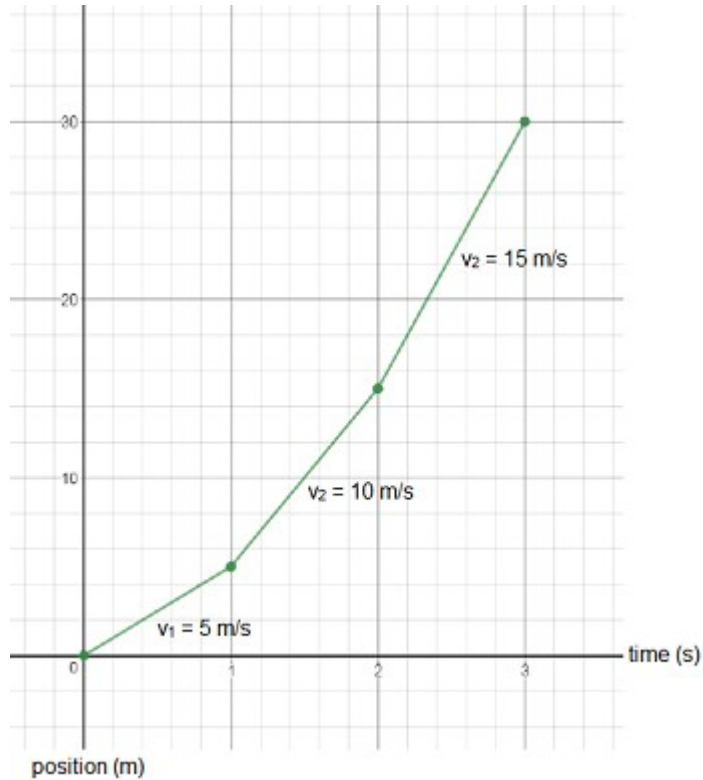
Conservation of angular momentum



Uniform and non-uniform motion



Uniform and non-uniform motion



Collision of particles

Collision is the process of exchange of momentum and energy between two particles.

Unlike the macroscopic view, two particles need not be in contact with each other for the collision.

Generally, collision is the result of interaction between the particles via some force e.g., Coulomb force.

In some cases, particles before and after the collision may not be identical, e.g., chemical reactions.

If the kinetic energy of the system, before and after the collision is conserved, collision is said to be elastic.

If the kinetic energy of the system, before and after the collision is not conserved, collision is said to be inelastic.

Elastic collision



Determine the final velocities of the two particles?

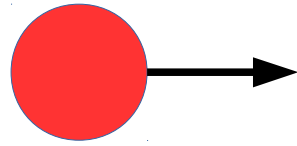
Elastic collision



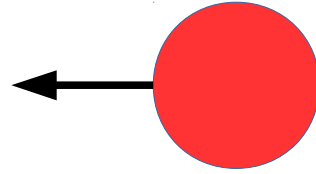
Determine the final velocities of the two particles?

- (1) $m_1 = m_2 = m$
- (2) $m_1 = m_2 = m$ and $\mathbf{u}_2 = 0$
- (3) $m_2 \gg m_1$
- (4) $m_1 \gg m_2$

Elastic collision



$m_1, \mathbf{u}_1$



$m_2, \mathbf{u}_2$

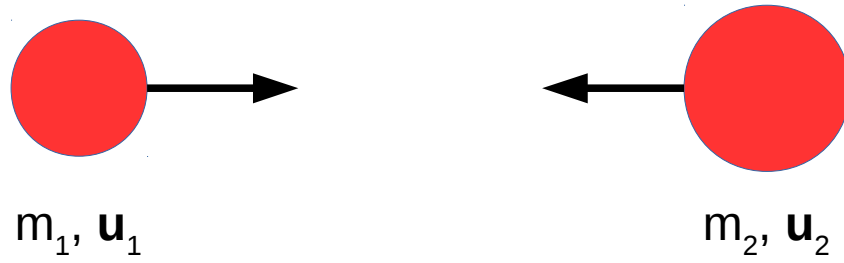
In center of mass frame of reference

$$\mathbf{u}_1' = \mathbf{u}_1 - \mathbf{V}_{\text{cm}}$$

$$\mathbf{u}_2' = \mathbf{u}_2 - \mathbf{V}_{\text{cm}}$$

$$\mathbf{V}_{\text{cm}} = (m_1 \mathbf{u}_1 + m_2 \mathbf{u}_2) / (m_1 + m_2)$$

Elastic collision



In center of mass frame of reference

$$\mathbf{u}_1' = \mathbf{u}_1 - \mathbf{V}_{\text{cm}}$$

$$\mathbf{u}_2' = \mathbf{u}_2 - \mathbf{V}_{\text{cm}}$$

$$\mathbf{V}_{\text{cm}} = (m_1 \mathbf{u}_1 + m_2 \mathbf{u}_2) / (m_1 + m_2)$$

$$\mathbf{p}_{u1} + \mathbf{p}_{u2} = 0$$

$$\mathbf{u}_1' = -(m_2 \mathbf{u}_2') / m_1$$

$$\mathbf{v}_1' = \mathbf{v}_1 - \mathbf{V}_{\text{cm-f}}$$

$$\mathbf{v}_2' = \mathbf{v}_2 - \mathbf{V}_{\text{cm-f}}$$

$$\mathbf{V}_{\text{cm-f}} = (m_1 \mathbf{v}_1 + m_2 \mathbf{v}_2) / (m_1 + m_2)$$

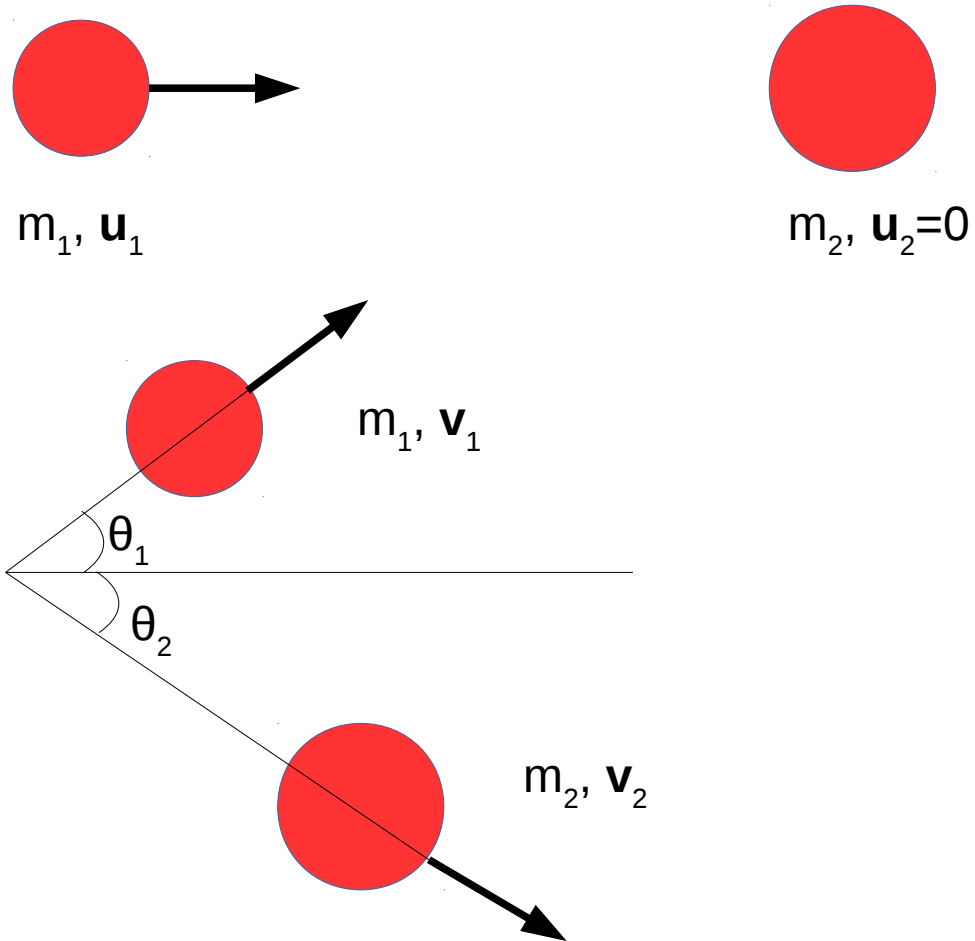
$$\mathbf{p}_{v1} + \mathbf{p}_{v2} = 0$$

$$\mathbf{v}_1' = -(m_2 \mathbf{v}_2') / m_1$$

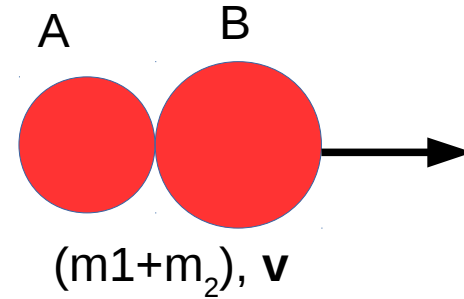
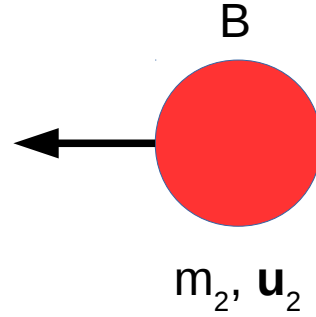
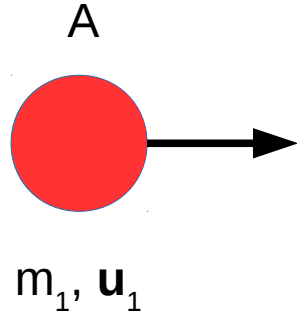
$$m_1 \mathbf{u}_1'^2 + m_2 \mathbf{u}_2'^2 = m_1 \mathbf{v}_1'^2 + m_2 \mathbf{v}_2'^2$$

$$\mathbf{v}_2'^2 = \mathbf{u}_2'^2$$

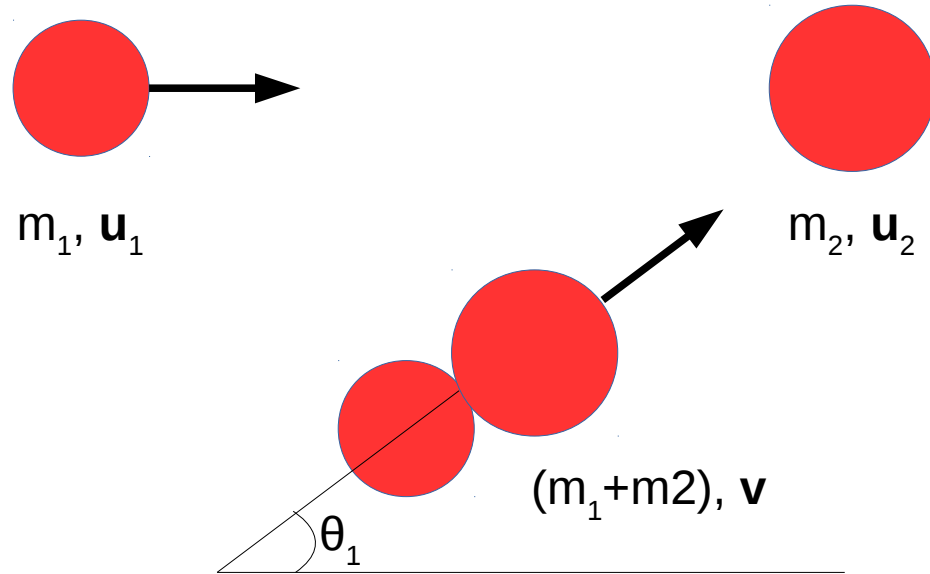
Elastic collision



Inelastic collision



Elastic collision



Inelastic collision



Determine the final velocities of the two particles?

Free fall : code

```
import numpy as np
import matplotlib as plt
import math

m = 10
d = 20
g = 9.81
t_max = (2*d/g)**0.5
t=np.arange(0,t_max,0.01)
h = d - 0.5*(g*t*t)
v = g*t
ke = 0.5*v*v*m
pe = m*g*h

plt.plot(t,h,"r", label="Hight")
plt.plot(t,v,"b", label="velocity")
plt.plot(t,ke,"y", label="kinetic energy")
plt.plot(t,pe,"k", label="potential energy")
Plt.legend()
plt.show()
```

Projectile motion: code

```
import numpy as np
import matplotlib.pyplot as plot
import math as m

v = 30
g = 9.8

theta = np.arange(m.pi/6, m.pi/3, m.pi/36)
t = np.linspace(0, 5, num=100)
for i in theta:
    x1 = []
    y1 = []
    for k in t:
        x = ((v*k)*np.cos(i))
        y = ((v*k)*np.sin(i))-((0.5*g)*(k**2))
        x1.append(x)
        y1.append(y)
    p = [i for i, j in enumerate(y1) if j < 0]
    for i in sorted(p, reverse = True):
        del x1[i]
        del y1[i]
    plot.plot(x1, y1)
plot.show()
```